

Summary Report: Candidate literary paper to be used

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1 Paper 1: Optimum government size and economic growth in case of Indian states: Evidence from panel threshold model

1.1 Summary

This paper investigates the relationship between optimum government size and economic growth using data of Indian states during 1990-91 to 2017-18. Our results derived from panel threshold regression model show a positive and significant impact of government size on economic growth within the estimated thresholds for both aggregate and sub-panels based on income and regions. Once the government size moves above the upper threshold level, then its impact declines and turns to be insignificant. Thus, our findings suggest the policymakers for maintaining the government size within the thresholds limit. Akram and Rath 2020

1.2 Data sources

- Indian states during 1990-91 to 2017-18
- Gross State Domestic Products (GSDP, 2011-12) and expenditures (such total expenditure, social sector expenditure
- revenue expenditure, and capital expenditure) data collected from Economic and Political Weekly Research Foundation (EPWRF).

1.3 Methodology

theoretical model which is proposed by Barro (1990, 1991).

$$Y(t) = AK(t)^\beta G(t)^{1-\beta}, \quad 0 < \alpha < 1$$

The thresholds model is as follows:

$$\delta y_{i,t} = \alpha) + \pi_i t + \beta_i y_{t-1} + \sum_{j=1}^k \psi_{i,j} \delta y_{i,t-1} + \epsilon_{i,t}$$

Table 5

Test for single/double/triple threshold. NSC stands for non-special category states. SC = denotes special-category states. LIG, MIG, and HIG stand for low-, middle-and high-income group states respectively. NR, SR, ER, and WR denote the northern, southern, eastern and western region respectively. GS denotes the government size. Results show the existence of two significant thresholds for full sample and sub-panels except for HIG. ***, ** and * denotes the significance level at 1%, 5% and 10%, respectively. p-values are given in the parenthesis.

Variables	Full Sample	NSC	SC	LIG	MIG
GS	<i>F</i> -stat	<i>F</i> -stat	<i>F</i> -stat	<i>F</i> -stat	<i>F</i> -stat
Test for single threshold	4.12 (0.67)	4.65 (0.65)	4.31 (0.41)	7.27 (0.44)	5.41 (0.60)
Test for double threshold	42.91*** (0.00)	41.83*** (0.00)	8.33** (0.03)	26.69* (0.06)	18.22* (0.08)
Test for triple threshold	2.49 (0.83)	3.77 (0.87)	4.52 (0.29)	3.84 (0.57)	4.54 (0.60)
GS	HIG	NR	SR	ER	WR
Test for single threshold	2.86 (0.62)	1.70 (0.90)	5.61 (0.65)	5.19 (0.51)	8.01 (0.17)
Test for double threshold	3.13 (0.59)	30.77*** (0.00)	43.07*** (0.01)	48.33*** (0.00)	19.15** (0.02)
Test for triple threshold	5.49 (0.54)	5.29 (0.26)	2.20 (0.80)	3.52 (0.88)	2.72 (0.48)

Table 6

Threshold estimate. NSC stands for non-special category states. SC = denotes special-category states. LIG, MIG, and HIG stand for low-, middle-and high-income group states respectively. NR, SR, ER, and WR denote the northern, southern, eastern and western region respectively. GS denotes the government size. γ_1' and γ_2' show the lower and upper thresholds. Results reported in this show the existence of thresholds for full sample and sub-panels except for HIG.

GS	Full	NSC	SC	LIG	IG	HIG	NR	SR	ER	WR
γ_1'	4.85	4.55	34.70	5.96	15.41	...	10.03	11.54	5.96	4.94
γ_2'	5.09	6.07	38.55	6.06	15.91	...	10.55	11.75	6.05	5.08

Figure 1: Caption for Table 1

1.4 Results

1.5 Conclusions

- **Objective:** The study addresses the literature gap regarding sub-national economic growth by examining the relationship between optimum government size and growth across major Indian states (1990–91 to 2017–18).
- **Methodology:** A panel threshold regression model was employed, identifying two significant thresholds for the aggregate panel and most sub-panels.
- **Optimal Thresholds (Aggregate):** Government size has a positive and significant effect on economic growth specifically when it lies between the lower and upper thresholds of 4.85% and 5.09%.
- **Category Variations:**
 - **NSC States:** Estimated optimal threshold range is 4.55%–6.07%.
 - **SC States:** Estimated optimal threshold range is significantly higher at 34.70%–38.55%.
- **Diminishing Returns:** Once government size exceeds the upper threshold, the coefficient of growth declines and becomes statistically insignificant.
- **Expenditure Composition:** Findings remain consistent when disaggregated into revenue, capital, and social sector expenditures, showing positive growth associations within threshold limits.
- **Policy Recommendations:** State governments should utilize fiscal rules and budgetary discipline to maintain an ideal size, ensuring that Keynesian deficit spending remains within the "optimum" range to avoid unfavorable growth consequences.

References

Akram, Vaseem and Badri Narayan Rath (2020). "Optimum government size and economic growth in case of Indian states: Evidence from panel threshold model". In: *Economic modelling* 88, pp. 151–162.